

Design Your Career On A VERY LARGE SCALE

VLSI field offers ample opportunities for professionals with subject-matter expertise in electronics, both in terms of challenging roles and monetary rewards. Here's how you can find place for yourself

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Are you ready for a 'technically challenging' career? If yes, get ready for something beyond the preliminary levels of integrated circuits (ICs) involving small-scale integration (like logic gates) or medium-scale integration (like multiplexers and parity encoders). A much bigger world involving miniaturisation at levels where even a micrometre or a micro-second is considered a huge measure is waiting for you! This is the world of very large-scale integration (VLSI).

"Programmable logic devices, hardware description languages and design tools of today have changed the circuit design process to such an extent that it has become totally independent of the actual device manufacturing issues," says Dr S. Karthik, engineering director, India Product Development Centre (IPDC), Analog Devices India. Let's figure out the possibilities and the opportunities in the VLSI field for professionals with subject-matter expertise in electronics.

Why VLSI?

In the present scenario, the outsourcing aspect of VLSI design is attracting major international players in this field to India. In fact, most of the top global players in VLSI have their R&D setups in India playing very pivotal role in driving their overall strategy.

Moreover, the electronics industry in India is undergoing a gradual but rapid change from a software-driven outsourcing model to one that is



increasingly focused on growth and development of the domestic market. According to India Semiconductor Association (ISA), the Indian semiconductor and embedded design industry is expected to earn revenues of \$43 billion in 2015—a marked increase from the \$5.9 billion in 2009. VLSI design services contribute approximately 13 per cent to overall revenues and 11 per cent to the overall workforce of the industry.

"Even in the recent downturn many companies did not cut the R&D spends. In fact, they are focused now, more than ever, on investing time and resources in creating the new applications for the wide varieties of market. Certain skill sets in this arena are al-

ways likely to be in short supply and therefore in great demand. So there is not only a need for talent in the semiconductor and electronics design industry but also a tremendous scope for growth for those who choose this path as a career," says Surinder Bhagat, country manager-HR, Freescale Semiconductor India.

Major recruiters in this field are Texas Instruments, PMC Sierra, Infineon, Alliance Semiconductor, Freescale Semiconductor, Analog Devices, Cadence, Synopsys, Mentor Graphics, HCL, Intel, Lucent, Micron Tech, National Semiconductor, Motorola, Philips Semiconductor, Qualcomm, Sasken, Atrenta, Conexant, Moschip, Cradle Tech, Synplicity, Wipro, TCS